

Oscillating droplet trains in microfluidic networks and their suppression in blood flow

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Droplets forming sequences in simple microfluidic networks are known to exhibit complex behaviours, but their dynamics are yet to be probed in channels long enough to accommodate many droplets simultaneously. Here we show that uniform sequences of liquid droplets flowing through microfluidic networks can spontaneously form ‘trains’ that periodically exchange between different branches of the network. Such system-wide oscillations do not rely on direct droplet–droplet interactions, are common to networks of various topologies, can be controlled or eliminated by adjusting network dimensions and can synchronize into larger flow patterns. The oscillations can also be suppressed via droplet collisions at diverging junctions. This mechanism may explain why red blood cells in microcapillaries exhibit only low-amplitude oscillations, preventing dangerous local hypertension or hypoxia that might otherwise ensue. Our findings are substantiated by a theoretical model that treats droplets as sets of moving points in one-dimensional ducts and captures the dynamics of large droplet ensembles without invoking the microscopic details of flows in or around the droplets. For blood flow, this simplified description offers more realistic estimates than continuous haemodynamic models, indicating the relevance of the discrete nature of blood to the excitation of oscillations.

The motions of cars over networks of roads, blood cells in networks of blood vessels or data packets along networks of cabled routers can often show collective dynamics that may synchronize into system-wide oscillations^{1,2}. While the emergence of such oscillations is still incompletely understood, it is known that they reflect not only system-specific factors (for example, driver behaviour, external regulation of arteries by muscles, route-selection computer algorithms) but also general features of the network, such as its topology, delays in flow related to channel length, common bottlenecks at merging junctions or competition between alternative paths (see refs. ^{3–6} for cars, refs. ^{7,8} for blood and refs. ^{9,10} for computer networks). Here, we study a related and important^{11–13} problem in the flow of droplets in ‘vascular’ microfluidic networks comprising long channels connected by diverging and converging junctions. We show experimentally, and confirm via modelling, that even with negligible interactions between neighbouring droplets, uniform droplet flows can spontaneously organize into long ‘trains’ that periodically exchange between a network’s various branches. Such oscillations are controlled by the network topology and dimensions, can be made to appear/disappear by reversing the flow direction and can synchronize into larger-scale flow patterns. Our models also rationalize (remarkably, for physiologically relevant parameters) the low amplitude of experimentally observed oscillations of blood flow in the microcirculation^{14–18} and ascribe it to the non-deterministic ‘turns’ red blood cells make at diverging capillary junctions. Overall, these results may have implications for the design of microfluidic systems directing droplet trains without the use of valves and also for the development of blood-replacement carriers¹⁹, determining physical parameters (particle size, concentration) that would ensure suppression of untoward oscillatory variations.

Motions of droplets in linear sequences, also known as single-lane flows^{20,21}, have been studied by many authors—both experimentally^{21–28} and also via simulations^{21–33}—but mostly in relatively simple arrangements of channels, such as two-channel loops^{21–30,33}.

Interestingly, even in such basic systems, the droplets were found to exhibit quite complex behaviours such as cyclic repetition of memorized sequences and transitions to chaotic dynamics for specific ranges of parameters^{21–26,29,30}. Studies of more intricate networks have been limited to cases with low occupation of channels and included series of simple loops³⁴, cascaded loops^{21,35} or large regular grids of short channels, mimicking porous materials³⁶. There have been no systematic studies of less trivial networks with channels long enough (as compared to their cross-section) to contain a large number (hundreds and more) of droplets.

In the present work, we consider networks of branching channels connecting a single inlet and a single outlet, which may be perceived either as a standalone system or a subsystem of a more general network. Droplets were always confined between channel walls and we required that their numbers and loads of resistance be large enough to let the droplets explore all branches (in this way, we ignore trivial cases of the pure filtering regime³⁷ in which all droplets follow the same path). The devices were fabricated in polycarbonate by milling and thermal bonding. In the experiments, channels accessible to droplets had a square cross-section ranging from 400×400 to 500×500 μm. To rule out any systematic dependence of the results on the method of droplet generation and/or small variations in droplet intervals and sizes, we generated droplet lane flows using three independent techniques: (1) a droplet-on-demand (DOD) system³⁸ based on electronically controlled valves; (2) syringe pumps; (3) free flow from a capillary-stabilized hydrostatic pressure head. The results we describe in the following are independent of these technical details.

Figure 1 (Supplementary Video 1) illustrates a typical experiment with a vascular-like network. Dye-coloured droplets select one of four equal-length paths between the network’s inlet and outlet. Initially, the droplets distribute uniformly over the network (Fig. 1a,b) but gradually organize into longer sequences that enter the same branch at the first bifurcation. Ultimately, the lengths of

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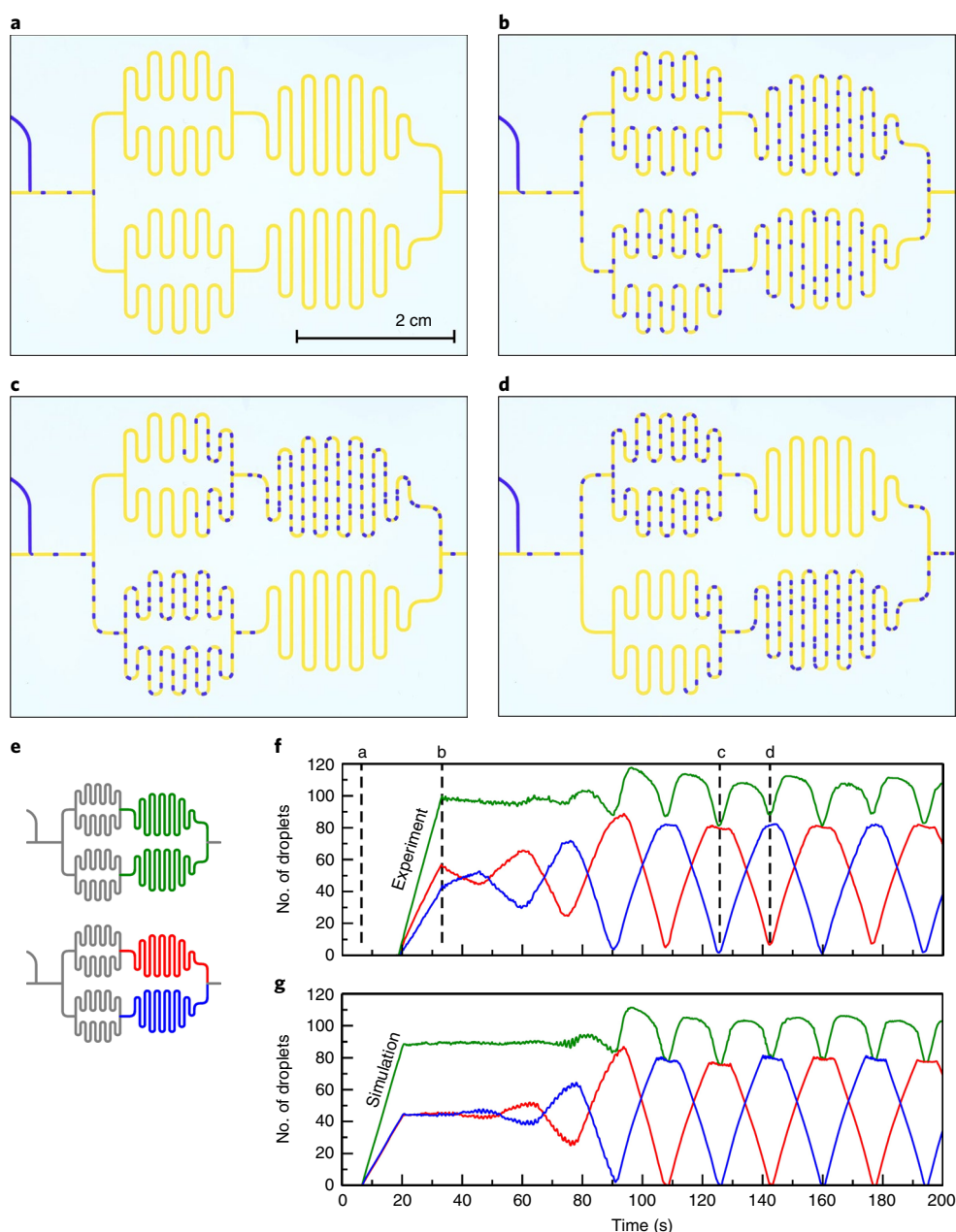


Fig. 1 | Emergence and quantification of large-scale oscillations in a simple network. Water droplets coloured with methyl blue and suspended in *n*-octane (containing 0.5 wt% Span-80 surfactant and 0.15 wt% Sudan I dye) flow into the network from left to right; see also Supplementary Video 1. **a**, The first droplet enters initial diverging bifurcation. **b**, When the same droplet passes the final converging node, other droplets trailing it are distributed over all of the network's branches. **c,d**, Subsequently, the oscillations emerge and droplet trains alternate between both main branches of the network. Note that due to volume preservation of both phases, the linear density of droplets in a train is approximately twice that of droplets distributing uniformly over the network. **e**, Coloured segments of the network used for counting droplets. **f**, Time dependence of experimentally observed numbers of droplets in these segments, with colours of plots matching the colours of segments in **e**. Times marked a, b, c, d correspond to the snapshots in **a–d**. **g**, An accompanying computer simulation started at $t=0$. The delay with respect to the experiment (position of point a) was chosen to match the phase of the emerging oscillations.

these sequences stabilize and they periodically alternate between the two main branches of the network (Fig. 1c,d). These oscillations are quantified in Fig. 1f and are reproduced in simulations in Fig. 1g (see later in the text).

A series of experiments and simulations with different network types allowed us to identify some regularities in the emergence and possible synchronization of droplet oscillations. First, the longest trains of droplets (that is, the largest non-uniformities in the distribution of droplets) are observed in networks in which the channels

diverging from the inlet are shorter than channels converging to the outlet. Interestingly, when the direction of flow in such networks is reversed, the oscillations are much less pronounced (see Fig. 3a–d later and Supplementary Videos 5 and 6). In extreme cases, the same network might exhibit oscillations when droplets flow from shorter to longer channels, but no oscillations in the opposite direction (Supplementary Fig. 1). Second, oscillations are observed over a wide range of fluid properties (for example, on varying the ratio of viscosity of both phases by three orders of magnitude or including

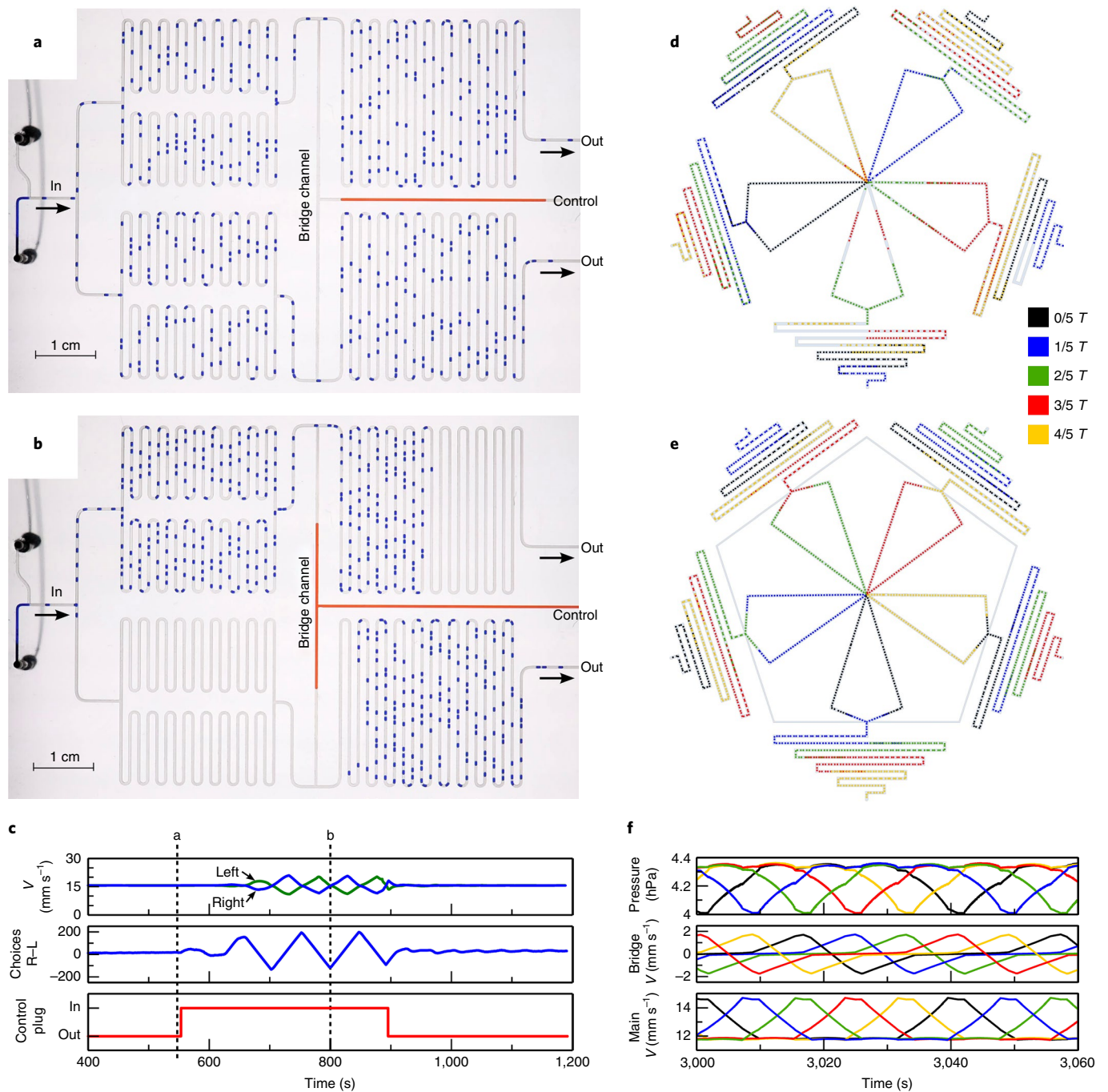


Fig. 2 | Regulation of flow patterns by side channels (bridges), non-accessible to droplets. a, b, Regulation of oscillatory versus uniform flow patterns by plugs of viscous liquid in the ‘bridge channel’ (see also Supplementary Video 3). Cross-sections of channels are 0.4×0.2 mm for the ‘bridge’ and 0.4×0.4 mm for all remaining channels, which—in the absence of the bridge—form a network similar to that in Fig. 1. The carrier liquid in the main channels is *n*-octane with 0.5% wt/wt of Span-80. The droplets are made of water dyed with methyl blue. The viscous plug is made of glycerin dyed with methyl red. **c,** Time evolution of the measurable flow characteristics (velocity of droplets in the initial sections of both branches and the difference in the total numbers of droplets that have entered the right (R) and left (L) branches) in response to the insertion and retraction of the viscous plug (bottom graph). Time instances a and b corresponding to the photographs in **a** and **b** are indicated by black dashed lines. **d, e,** Simulations (cf. Supplementary Video 4) in which droplets flow outwards from a central inlet through pairs of outward channels (‘oscillators’) towards five outlets kept at the same pressure. Each image is a superposition of five different snapshots separated by intervals of $1/5 T$ (where T is the period of a given oscillator in a stationary state). Droplets are colour-coded according to these times. In the simulations, the cross-sections of the bridges are 0.4×0.2 mm and those of all other channels are 0.4×0.4 mm. **f,** Further quantification of the simulations (top to bottom): pressures of internal nodes in the stationary state and the time dependencies of velocities in the bridging and main channels.

systems of gas bubbles in a liquid phase; Supplementary Fig. 2). Third, the oscillations are accompanied by rhythmic variations of pressures and flow rates between the network’s branches. This is viv-

idly illustrated by the system in Fig. 2a–c, where two branches of the network are connected by a transverse channel. When this channel is ‘open’ (that is, filled by the same liquid as the rest of the network),

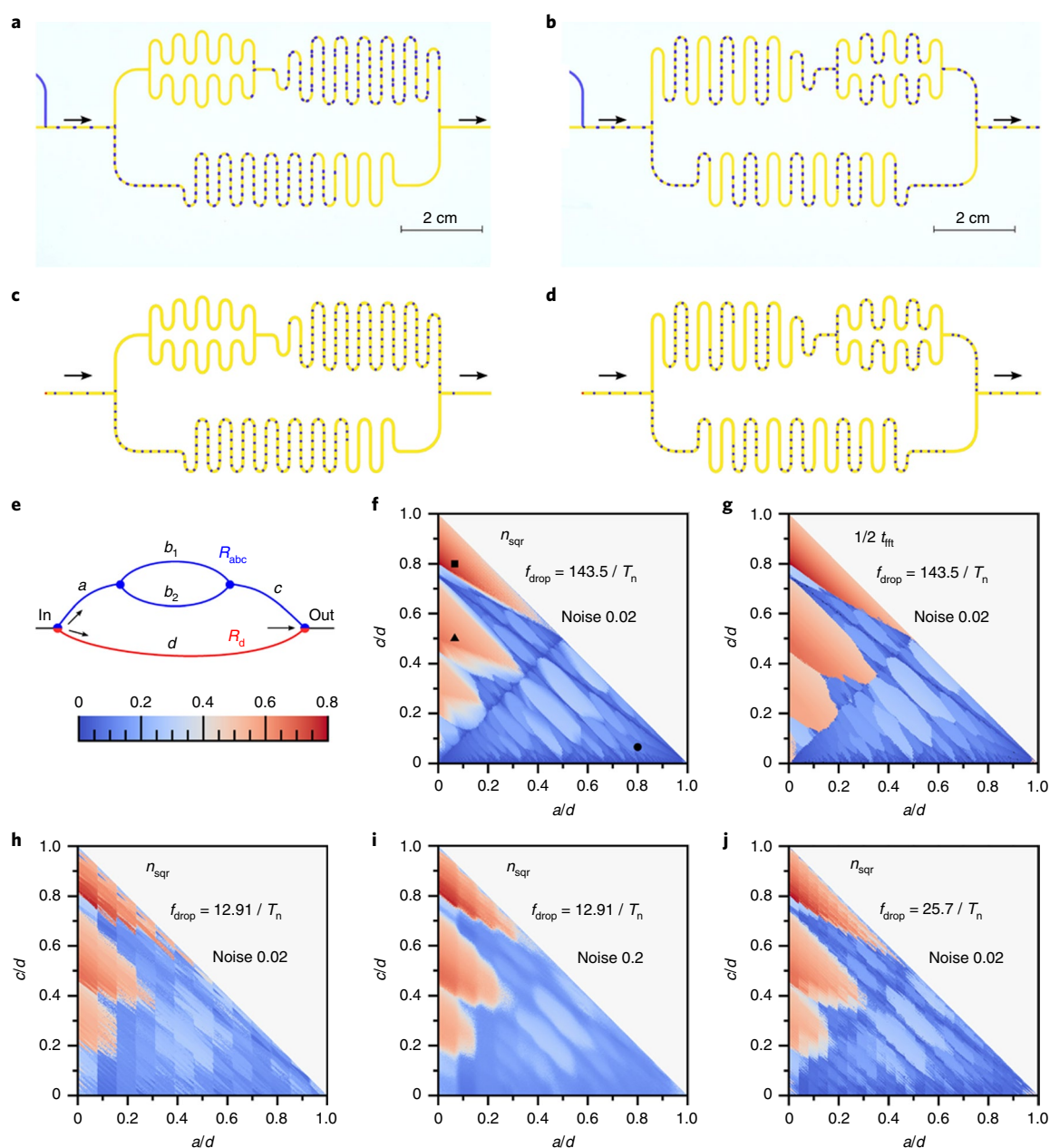


Fig. 3 | Experiments and simulations of the simplest oscillating network. **a**, An experimental snapshot of stationary oscillations in a network comprising parallel connection of two channels where one contains an internal loop. Droplets of water (dyed with methyl blue) flow in *n*-hexadecane (with 0.5 wt% Span-80 and 0.15 wt% Sudan I yellow dye). **b**, The same chip with reversed direction of flow (mirrored photograph). **c,d**, Simulations of systems from **a** and **b**, respectively. See also Supplementary Videos 5 and 6, which compare these experiments to simulations. **e**, Generalization of the network with the topology as in **a–d**. The colour scale quantifies the amplitude of oscillations used in the parameter maps in **f–j**, with different parameters of the incoming stream of droplets. The triangular shape of the maps results from the fact that pre-balancing is possible only when $a + c < d = 1$ (for discussion of pre-balanced versus unbalanced networks, see Supplementary Sections 3.2 and 3.4). Each map is based on over 120,000 simulations (500 pixels along each axis). All simulations were performed for a droplet's resistive load of $0.006R_d$ (meaning that the presence of a single droplet in channel d increases its hydraulic resistance from R_d to $1.006R_d$, modulo emulated noise). Other parameters of the stream of droplets—the frequency (f_{drop}) and the amplitude of emulated noise (both δ_i and δ_o) are explicitly stated in each map. **f,g**, Maps related to the same system described by two different quantities: n_{sqr} (**f**) and half-period ($T_{fit}/2$) (**g**). If corresponding pixels of these two maps have the same colour (which is true for more than half of the points), the oscillations are perfect, that is, formed by exactly two trains per cycle, one flowing in branch abc and the other in d . Additional symbols in the map in **f** correspond to experimental systems shown elsewhere: Filled square: system shown in **a**, Supplementary Section 2, Supplementary Fig. 2, Supplementary Videos 5 and 9 ($a = 0.066$, $c = 0.8$); filled circle: system in **b**, Fig. 4d and Supplementary Video 6 ($a = 0.8$, $c = 0.066$); filled triangle: system in Fig. 4a–c, Supplementary Section 5 and Supplementary Fig. 8 ($a = 0.066$, $c = 0.5$). **h–j**, Maps plotting the amplitude (n_{sqr}) for networks with two smaller f_{drop} and two different noise amplitudes.

its low resistance allows the pressure between the two branches to equalize at all times, resulting in uniform droplet flows. When, however, the bridge is partially filled by a plug of a very viscous

liquid, its resistance becomes high enough such that oscillations in the main channels emerge (as in the system without the bridge, Fig. 1). The back-and-forth motions of the plug are clearly visible

in Supplementary Video 3 and reflect small (-100 – 100 Pa), periodic changes of pressure difference between the ends of the control channel. Fourth, in more complex networks, ‘communication’ between different sub-branches—via transverse channels transmitting pressure but not droplets—can synchronize the droplets into larger-scale spatiotemporal patterns. In this spirit, Fig. 2d–f and Supplementary Video 4 show a network in which droplets flow from a central node into five radially oriented loops, all connected to the same-pressure outlet. If the intermediate nodes are not bridged, droplet trains in each loop emerge but are not ‘globally’ synchronized. When, however, the intermediate nodes are connected by transverse bridges, droplet sequences gradually synchronize and ultimately develop rotary patterns over the radial channels (Fig. 2e and Supplementary Video 4) with the symmetry-breaking into clockwise or anticlockwise directions being, as expected, random. Fifth, we have established that the simplest network supporting oscillations contains two branches, with one splitting and joining back to form a loop (Fig. 3).

To explain these observations, we adapted a simplified model²⁰ that treats droplets as sets of moving points in one-dimensional ducts and captures the dynamics of large droplet ensembles without dealing with the microscopic details of flows inside or in the neighbourhood of droplets (note that volume of fluid (VOF)³⁹ and boundary element methods³³ are capable of modelling interfaces of flowing, colliding, coalescing or splitting droplets, but require excessive computational power and are largely redundant in explaining the phenomena we describe). In our model, the flow rate Q_i in the i th channel is calculated as $Q_i = \Delta p_i / R_i$, where Δp_i is the difference in pressures between the nodes at the ends of this channel, and R_i represents the hydraulic resistance to flow. Importantly, both nodal pressures and flow rates in the channels may be easily calculated by solving a system of linear equations, similar to the Ohm and Kirchhoff laws of electric circuits. Droplets then move along channels with superficial flow velocity $V_i = \beta Q_i / A$, where β is the so-called mobility factor⁴⁰ (assumed here, due to the square cross-section of the channels, to be $\beta = 1$) and A is the cross-sectional area (here assumed constant). At diverging junctions, droplets turn into channels with the highest flow rate. At this level of description there is nothing non-trivial in the dynamics of such droplets—they will simply follow the optimal path. However, the presence of droplets in itself increases the hydraulic resistance of the channel in which they flow, so for n identical droplets $R_i = R_i^0 + nr$, where R_i^0 represents the hydraulic resistance of the i th channel without droplets and r is the contribution of an individual droplet to this resistance (assumed to be additive and independent of a droplet’s position or distance to neighbouring droplets⁴¹). In particular, droplets that follow a given path may elevate its resistance so much that subsequent ones must choose another route. Even though the model does not account for direct interactions between neighbouring droplets (which is justified as long as the gaps between them are larger than the droplet diameters), the ‘point loads of resistance’ carried by the droplets establish system-wide droplet–droplet dependencies. Such global effects are evidenced by the resistances, flow rates and nodal pressures evolving with the motion of every single droplet from channel to channel. Finally, to account for the imperfectness in manufacturing and the inherent fluctuations of the parameters, we emulated artificial noise, as detailed in Supplementary Section 3. The role of noise in simulations of the dynamics of droplets in microfluidic networks has been studied in greater detail in ref. ⁴².

The model thus formulated reproduces faithfully the emergence of oscillations in networks of different dimensions and connectivities (Figs. 1f,g and 3a–d and Supplementary Videos 5 and 6) and enables a systematic study of the parameters involved—here, the results are illustrated for the simplest network topology for which oscillations are observed (Fig. 3a–e). The requirement that all three paths traversing this network must be accessible even to the first droplet (we call it pre-balancing; Supplementary Section 3.2) reduces the

number of independent parameters to two. Choosing these parameters as a and c , key flow characteristics can be captured in triangular maps (Fig. 3e) plotting calculated amplitudes/lengths and periods of the droplet trains (n_{sq} and t_{int} , respectively; for formal definitions see Supplementary Section 3.3). The key observation from all maps shown in Fig. 3f–j is that oscillations are quite ubiquitous and not limited to small regions of parameters. The maps also confirm the experimental result that oscillations are more pronounced when channel a between two diverging junctions is shorter than channel c between two converging junctions. Even in this regime, however, there are combinations of parameters for which oscillations are weak or non-existent. For example, in the maps in Fig. 3f,g there are no oscillations along the lines defined by equations $a + c = 0$ or $3a + 4c - 3 = 0$. For these parameters, the choices made by the droplets at the junctions are akin to coin flipping, although we note that they have different statistics. Furthermore, as long as the number of droplets residing in the network at a given time is large enough, the maps do not change strongly when the resistive loads of the droplets, the amplitude of the noise or the frequencies of dripping (that is, linear densities of droplets, f_{drop}) are varied (Fig. 3h–j and Supplementary Section 5). Finally, we confirmed that oscillations are present in many (we tested hundreds) other, more complex networks. In particular, the simulations detailed in Supplementary Section 6 and Supplementary Video 7 evidence that all combinatorial rewirings of the innermost channels between diverging and converging ‘trees’ (rendering the network non-planar) also exhibit system-wide oscillations, despite the fact that equations describing them become much more complicated. These results indicate that our model can be applicable to the study of non-planar networks such as those often found in natural systems.

In this context, one particularly interesting problem comprises the systematic (0.07 – 0.13 Hz) but low-amplitude (up to $\pm 20\%$ of average flux) oscillations of red blood cells (RBCs)^{16,18}, which have been demonstrated experimentally in systems mimicking the microcirculation *in vitro*¹⁷ in animals with functioning heart and lungs¹⁶ and in humans during non-pulsatile cardiopulmonary bypass¹⁸—a condition that eliminates a potential relation between the observed regular oscillations of the haematocrit and rhythmic operation of these organs. The mechanisms of these self-sustaining oscillations are not completely understood and—even excluding the abovementioned rhythms of large organs—several physiological explanations may apply^{14,18}. Although some haemodynamic models of blood flow predict the emergence of oscillations in certain regimes^{7,8}, the values of key parameters required for these oscillations are not realistic for real blood (see the discussion below).

Given that—akin to our droplets—RBCs flow in single file⁴³, are strictly confined/deformed by the walls of the surrounding microvessels, and increase these vessels’ resistance to flow, the small amplitude of their oscillations might be surprising (although certainly beneficial from a physiological point of view by, for example, preventing local hypoxia or hypertension). We suggest that the reason for such small amplitudes might lie in the fact that the RBCs at the junction of arterioles do not necessarily follow the paths of least fluidic resistance. In particular, collisions between two or more RBCs can cause them to move into a channel of lower flow rate, an effect that we actually reproduced in experiments with closely spaced droplets colliding at junctions. As illustrated in Fig. 4 (and with further discussion in Supplementary Section 7), this phenomenon is fully deterministic; as long as intervals between incoming droplets are controlled, it is possible to enforce the transition from fully developed oscillations (as in Fig. 4b) to an entirely uniform distribution of droplets (Fig. 4c and Supplementary Video 9) and vice versa (Supplementary Video 9), even without changing the average density of droplets or average intervals between them. Beyond these extremes, all intermediate distributions of droplets can be arranged by purposefully placing closely separated droplets

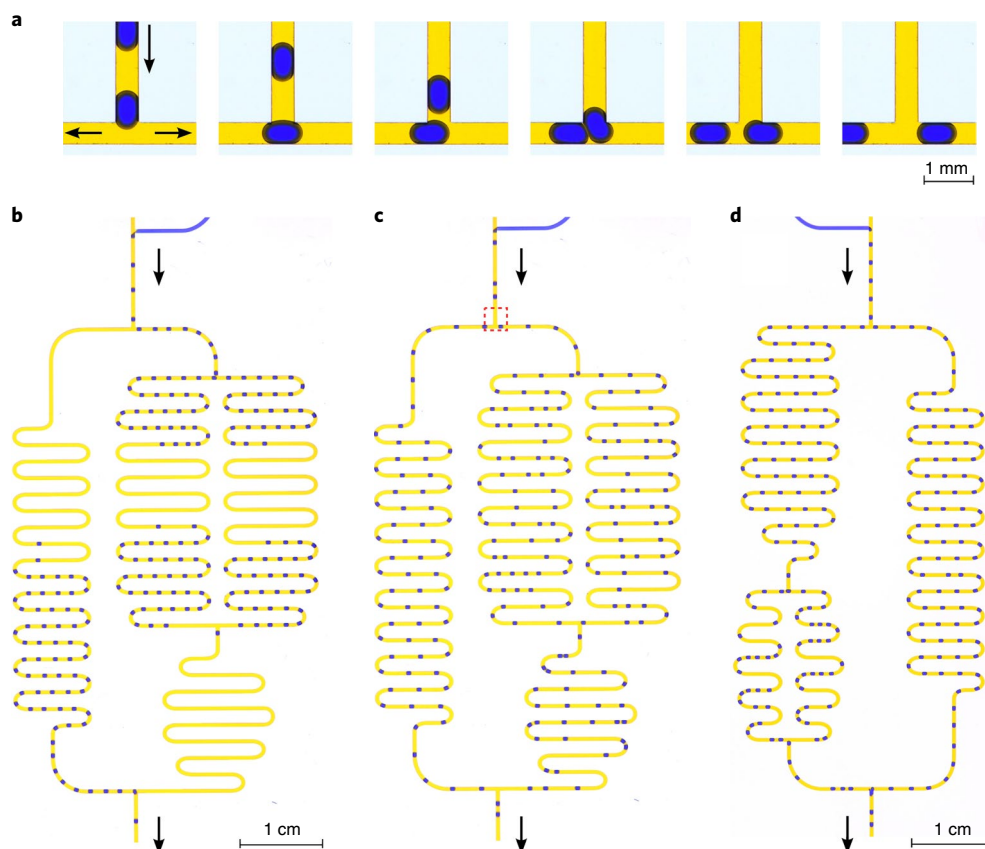


Fig. 4 | Suppressing and avoiding oscillations by collision-driven dynamics of droplets. **a**, Mutual collision and scattering of two droplets arriving at a T-junction one after the other within a short interval. Droplets are 0.8 mm long and their mutual distance (centre to centre) is 1.6 mm at the first image. Subsequent snapshots are taken with a constant time interval of 1/12 s. As a result of the collision, the second droplet is pushed into the branch with a lower flow rate. For further discussion of these collisions, see Supplementary Section 7 and Supplementary Video 8. **b**, Experimental snapshot of stationary oscillations in a network indicated by the filled triangle in Fig. 3f. Droplets are produced by a DOD valve with two alternating intervals (3.0 and 3.6 mm). **c**, The same network and the same average distance between incoming droplets, but the two alternating intervals are 1.6 and 5.0 mm. As a result of collisions at the initial T-junction (red border around this junction corresponds to **a**, where one such collision is shown) the oscillations do not emerge, or are suppressed shortly after transition from the oscillating state shown in **b**. **d**, The network of Fig. 3b, with droplets generated at constant intervals of 2.2 mm (centre to centre)—short enough to maintain the non-oscillating state due to mutual collisions, but not short enough to suppress oscillations if the experiment were started from an oscillating state.

among longer intervals in the incoming stream (Supplementary Section 7). Even for constant intervals, if they are short enough, collisions may prevent development of the oscillations, as in Fig. 4d, although this mode of flow is metastable (applying the same intervals to the system already in oscillatory dynamics would not be enough to quench it).

Collisions of RBCs in blood are much less deterministic. First, due to the flattened shape of an RBC, the process of passing a junction depends on the initial orientation of the RBC, which introduces some randomness, even at the level of single cells. Second, due to the high concentration and elevated stiffness, mutual collisions at areas of bifurcation may take longer and involve more than two RBCs, leading to flow obstruction by temporary structures^{44,45}. This intrinsic (or effective—as it results from the lack of complete knowledge) indeterminacy can be accounted for in our model by introducing the so-called ‘plasma skimming function’^{7,8} (Fig. 5a). This function, widely used in haemodynamics, quantifies ‘how deterministic’ is the division of the haematocrit between two branching arterioles with volumetric flows Q_1 and Q_2 . The probabilities of RBCs entering these two channels are, respectively, $Q_1^s / (Q_1^s + Q_2^s)$ and $Q_2^s / (Q_1^s + Q_2^s)$, where parameter s controls the level of determinism. For $s \rightarrow \infty$ the choice is fully

deterministic, and droplets always enter the channel with the largest flow rate, as in our experiments described above (provided that there are no collisions). For $s = 1$, the probabilities of choosing a channel are proportional to the flow rates, and for $s = 0$, they are equal regardless of the flow rates (which does not make physical sense). Mathematical models of blood microcirculation based on in vivo data⁴³ estimate that for bifurcations at the smallest capillaries $s = 2$ –2.5. At the same time, studies based on continuous haemodynamic models report that oscillations are observed only at $s > 5$ (ref. 7) or $s > 4$ (ref. 8; where the threshold s could be lowered at the expense of an unrealistically large exponent in the nonlinear dependence of apparent blood viscosity on the haematocrit). Simulations for our networks quantified in Supplementary Fig. 11 show a general trend (holding for different parameter values) that as s decreases—that is, when choices at junctions become more random—the oscillations indeed fade away. Importantly, with the network from Fig. 1 rescaled to the size typical for the blood microcirculation (Fig. 5b) and with parameters f_{drop} and r chosen to mimic the rheological properties of blood (Supplementary Section 9), clearly visible oscillations are observed for s as small as 2.2 (Fig. 5c,d)—that is, for the value of s matching the in vivo data (cf. above).

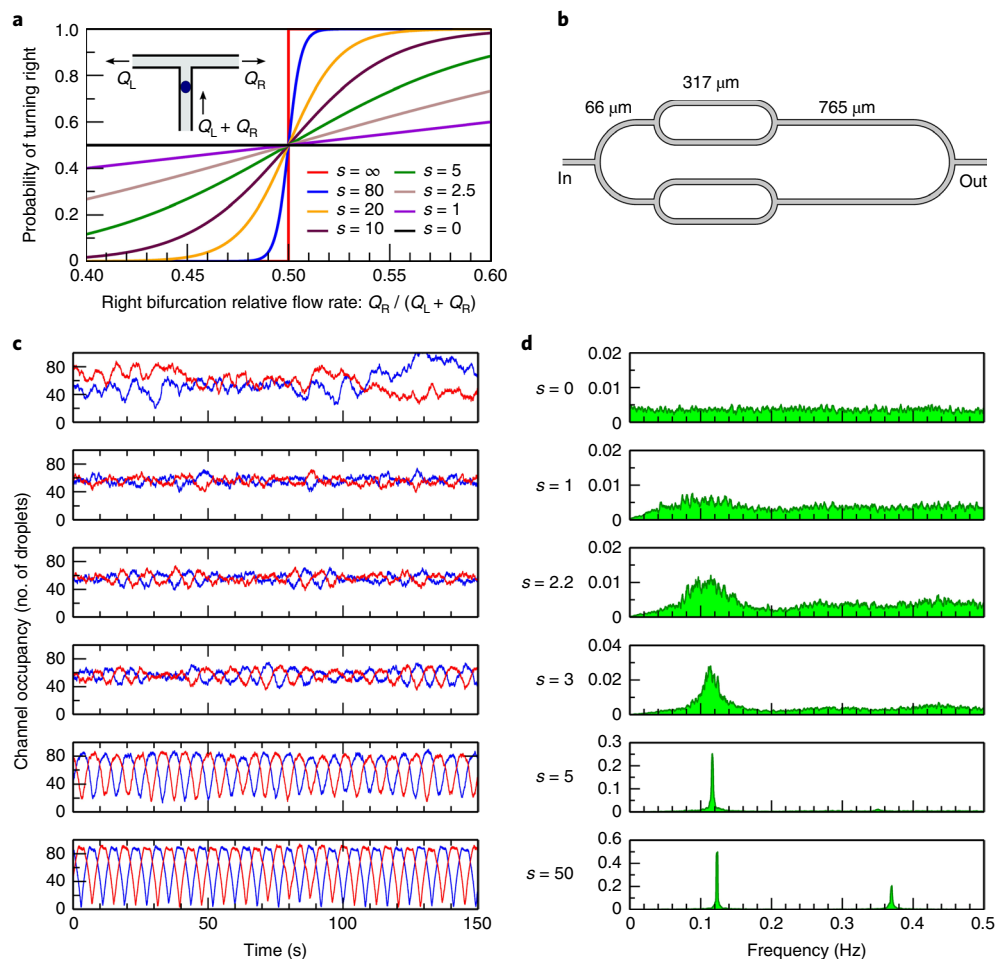


Fig. 5 | Non-deterministic turns of droplets (or blood cells) at diverging junctions and oscillations in networks rescaled to mimic blood microcirculation.

a, Plasma-skimming function mimicking the non-deterministic choice of branches at diverging junctions, plotted for different values of plasma-skimming parameter s . For infinite s , the model becomes fully deterministic—droplets follow the largest flow rate. **b**, Network with topology as in Fig. 1 but rescaled to mimic blood circulation, with diameters of capillaries set to $6\ \mu\text{m}$. Flow rates and assumed concentrations of RBCs are detailed in Supplementary Section 9. **c**, Graphs counting the number of RBCs in the two longest channels as a function of time (colours as in Fig. 1e). The origin of the time axis corresponds to the stationary state established during simulation. **d**, Fourier transforms of the time-series H_i (defined as +1 if the i th droplet arriving at the first bifurcation turns left, or -1 if right; Supplementary Section 3.3). The parameters of the simulations—that is, plasma-skimming exponents, s —are given between corresponding panels. For $s=1$, when the division of RBCs at the branching channels matches the division of flow rates, the Fourier spectrum of H_i features only a flat maximum at a frequency reciprocal to the sum of flow times through both branches of the network (that is, lower than the frequency of oscillation for larger s). At around $s=2.2$, the oscillations at $\sim 0.1\text{ Hz}$ become easily visible both in the time and frequency domains; in the latter, the peak amplitude is roughly three times larger than the flat part of the spectrum (representing the level of white noise) and six times larger than the minimum of the spectral density to the right of the peak.

The frequency of these oscillations is 0.1 Hz , which also matches the frequencies observed *in vivo*^{16,18}.

The fact that our simplified, discrete model gives more realistic estimates than sophisticated continuous haemodynamic models points to an intriguing conclusion that the discrete nature of RBCs may play a more important role in haematocrit oscillations than many subtle interactions and nonlinearities underlying continuous descriptions (for further discussion, including the consequences of the discreteness of RBCs, see Supplementary Section 9).

We envision three possible developments from this work: (1) in further studies of the mechanism of oscillations and their suppression in blood, incorporating alternative or coexisting mechanisms (for example, the role of leukocytes¹⁷, varying lumen of vessels⁴² and their active control¹⁴⁶); (2) the design of microfluidic circuits in which the network topology alone would direct the droplet cargos

to desired locations without the use of any valves (cf. the example in Fig. 2d–f); (3) the development of blood substitutes¹⁹, especially in estimating the concentrations and sizes of oxygen-carrying particles ensuring that they undergo collisions at capillary junctions, thus preventing large-scale oscillations.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding authors upon request.

Code availability

The computer codes used for simulations are available from the corresponding authors upon request.

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Author contributions

O.C. conceived the idea and performed experiments and simulations. P.G. and B.A.G. supervised and guided the research. All authors participated in the writing of the manuscript.

Competing interests

The authors declare no competing interests.

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